Comparing Statistical Diagrams

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 27 January 2026

Comparing statistical diagrams

How do I compare statistical diagrams?

- You may be given **two graphs** for two different **data sets** with the same **context**
- Compare **trends** in the graphs
  - **Increases**, **decreases**, **maximum points**
  - **Steepness** of the change
- Comment on **differences** and **similarities**
- Explain **clearly** which **part** of the graph you are talking about
- Use **numbers** from each data set in your comparisons
- Comment on any **extreme values** (outliers)
- Calculate **means, medians** or **modes** to compare **averages**
- Calculate **ranges** to compare the **spread** of values

Examiner Tips and Tricks

Always relate numbers and calculations back to the context of the question.

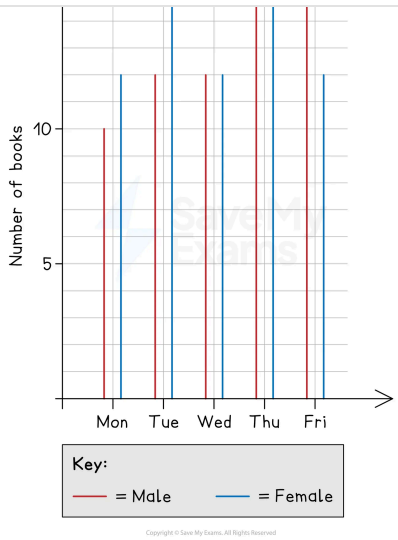
If you don't, you won't score all the marks!

Help me understand

Worked Example

The number of books bought during the opening week of a new bookshop is shown below.
The shopkeeper wants to investigate shopping patterns between male and female customers.

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



(a) Briefly summarise the overall trend in the number of books bought by male customers.

Answer:

Look at the heights of the red lines
Comment on how they increase ('briefly' means you do not have to talk about every point)
Use numbers from the graph and exact phrases from the question

The number of books bought by male customers increases from 10 on Monday up to 15 on both Thursday and Friday

(b) The shopkeeper says that the changes in the blue lines suggest that the amount of books bought by females varies more than males.

By comparing a suitable statistical measure, show that this is not necessarily true.

Answer:

Variations refers to the spread of data, so compare their ranges
For male and female, subtract the smallest value from the biggest value

Male range is 15 - 10 = 5

Female range is 16 - 12 = 4

Compare the numbers
Explain how this relates to the spread of the data
State how your findings do not agree with the shopkeeper

The range of male customers is greater than that of female customers, 5 > 4
This suggests male customers have a greater spread (variability) in books bought
This is the opposite of what the shopkeeper said

(c) Give one reason as to why the shopkeeper should not use the data shown to draw conclusions about shopping patterns.

This data is for the opening week of the bookshop only
State that this is unrepresentative of a normal week
Give a specific real life example

The data shown is for the opening week of the bookshop
It is unlikely to be representative of a normal week
The numbers may increase as the bookshop becomes more popular

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.